
Are We Solving the Right Problems? Building AI That Matters for Radiotherapy

Stine Korreman

Professor of Medical Physics

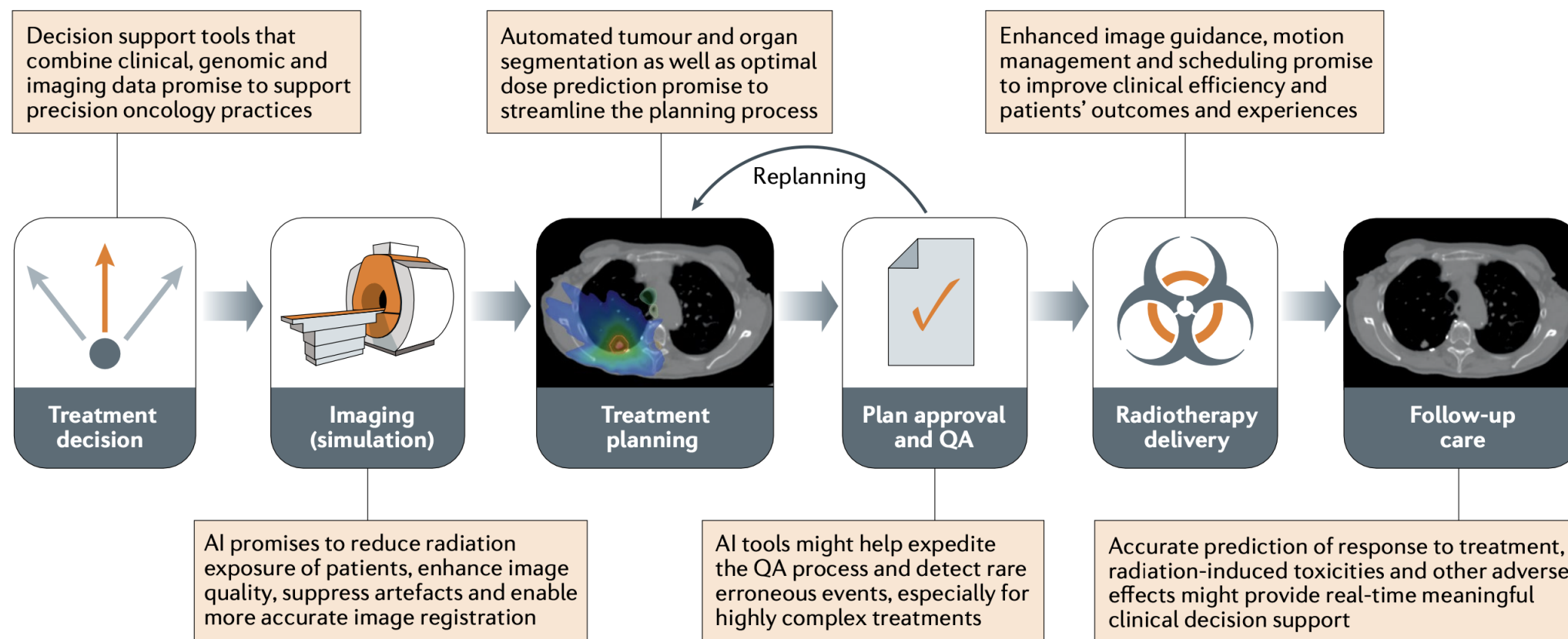
Director of AIM@CANCER research center

Danish center for Particle Therapy

Faculty of Health Sciences, Aarhus University



Artificial Intelligence can play a role in most parts of the radiotherapy chain



We envision a fully AI augmented radiotherapy clinic for the future

Smart patient scheduling



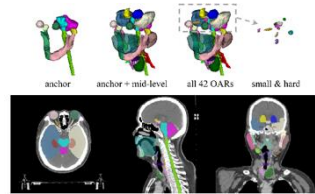
Shared decision making



Image acquisition



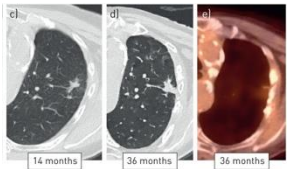
Image segmentation



Quality assurance analysis



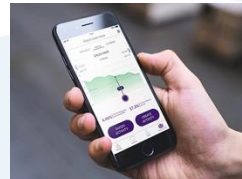
Response monitoring



Patient monitoring



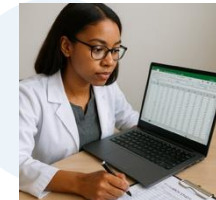
PROM collection



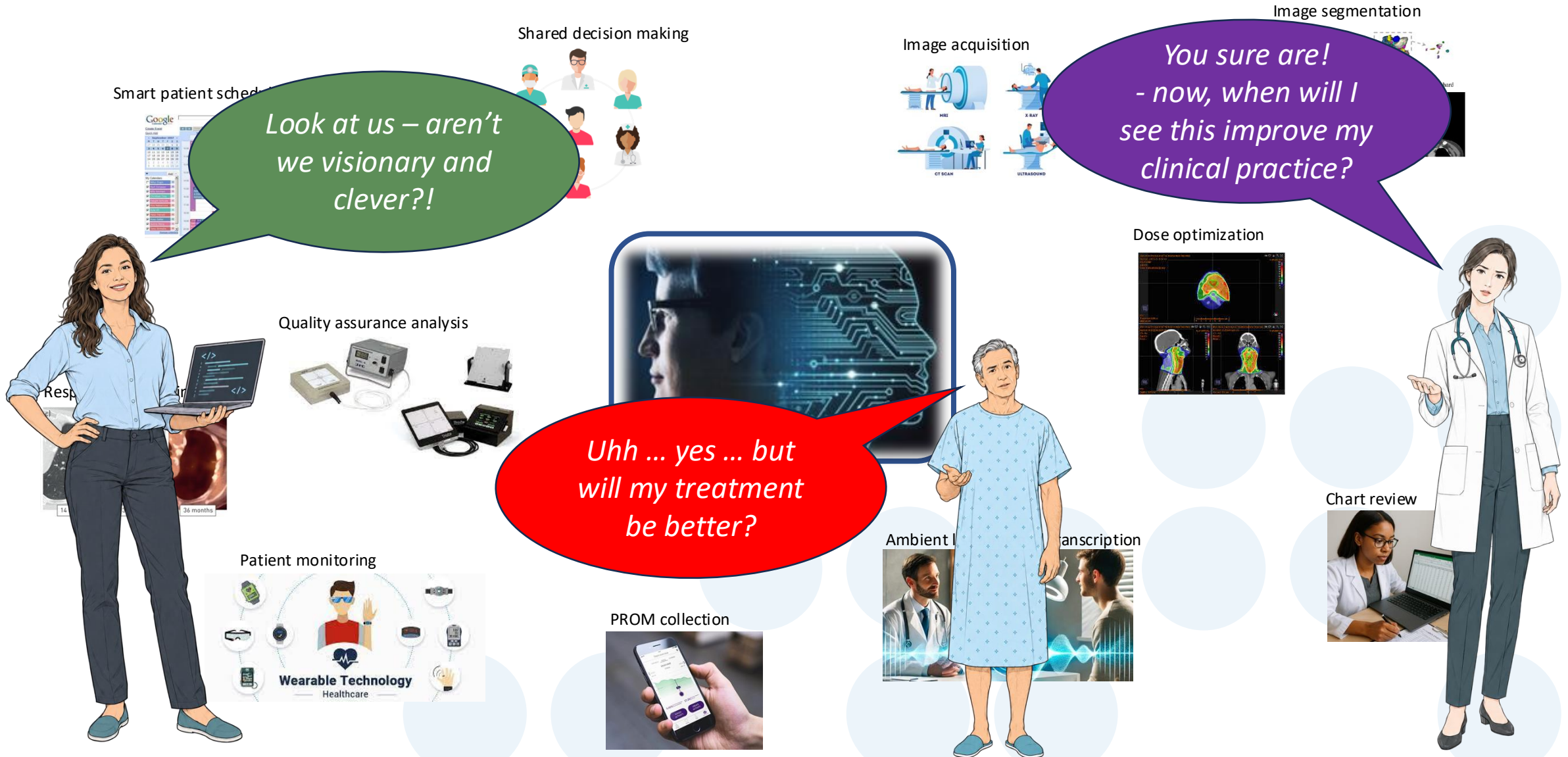
Ambient listening and transcription



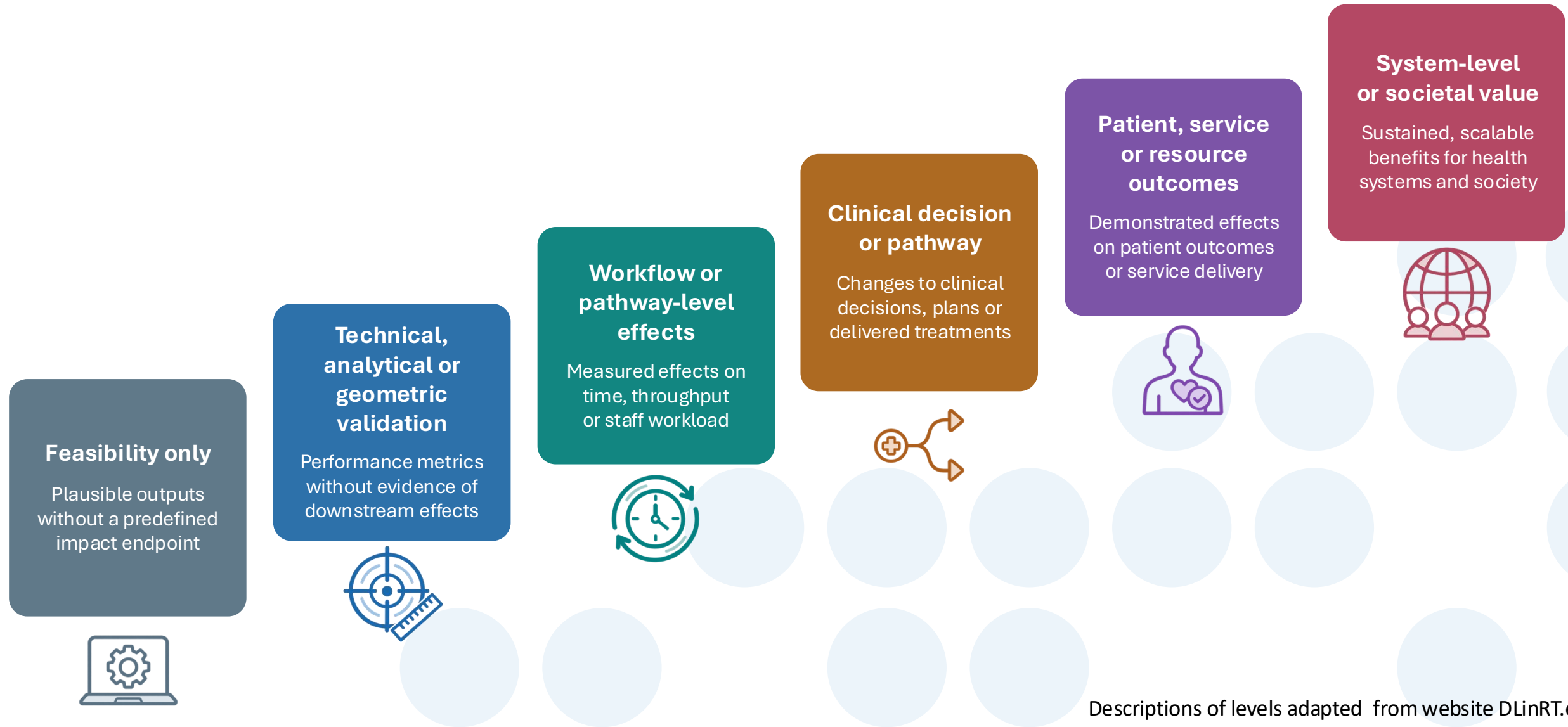
Chart review



We envision a fully AI augmented radiotherapy clinic for the future



What categories of impact do we expect from AI in a clinical context?



Example: Multiple studies confirm the ability of deep learning models to segment target volumes

npj | digital medicine

Published in partnership with Seoul National University Bundang Hospital

Multicentre evaluation of deep learning-based automatic segmentation of head and neck target volumes for radiotherapy

Eric Pei Ping Pang^{1,2,15}, Hong Qi Tan^{1,2,3,15}, Susan Ramadan⁶, Timo Kiljunen⁸, Marta C. Ingrid Kiitama⁸, Danil Zolotuhhin⁸, Eduard Jani Keyriläinen^{11,12,13}, Miia Mokka¹¹ & Me

> Cancer Imaging. 2022

Automatic segmentation of head and neck tumors on MRI

Jens P E Schouten¹, S C René Leemans³, Pin

Article

Head and Neck Gross Tumor Volume Automatic Segmentation Using PocketNet

Awj Twam¹, A

PMCID: PMC1

ACTA ONCOLOGICA
<https://doi.org/10.1080/0284186X.2021.1949034>

ORIGINAL ARTICLE

Comparing different CT, PET and MRI multi-modality image combinations for deep learning-based head and neck tumor segmentation

Jintao Ren^{a,b,c}, Jesper Grau Eriksen^{a,d}, Jasper Nijkamp^{a,b,*} and Stine Sofia Korreman^{a,b,c,*}

^aDepartment of Clinical Medicine, Aarhus University, Aarhus, Denmark; ^bDanish Centre for Particle Therapy, Aarhus University Hospital, Aarhus, Denmark; ^cDepartment of Oncology, Aarhus University Hospital, Aarhus, Denmark; ^dDepartment of Experimental Clinical Oncology, Aarhus University Hospital, Aarhus, Denmark

Radiotherapy and Oncology 182 (2023) 109574

Contents lists available at ScienceDirect

Volume Segmentation in Head and Neck Cancer

Authors: Xin He, WeiJie Chen, Zachary Huemann, Brayden Schott, Nuohao Liu, Tyler J.

Challenge, HNTS-MRG 2024, Held in

PET/CT T1w T2w

DSC(GTVt/n): 0.78/0.87

Ground truth Prediction

(c) (f)

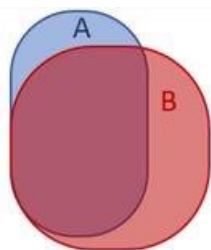
Most literature on AI in RT reports technical evaluation of quantitative metrics

Measures of benefit

Volumetric Dice Similarity Coefficient

$$\frac{2(A \cap B)}{A + B}$$

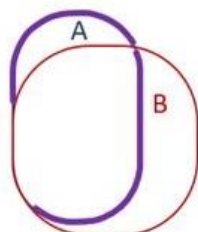
where A and B are volumes



Added Path Length (pixels)

$$A - (A \cap B)$$

where A is the manually corrected segmentation surface and B is the autosegmentation surface

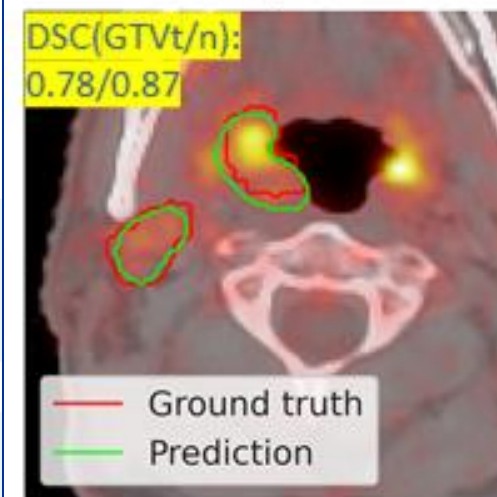


Time measurement:



PET/CT

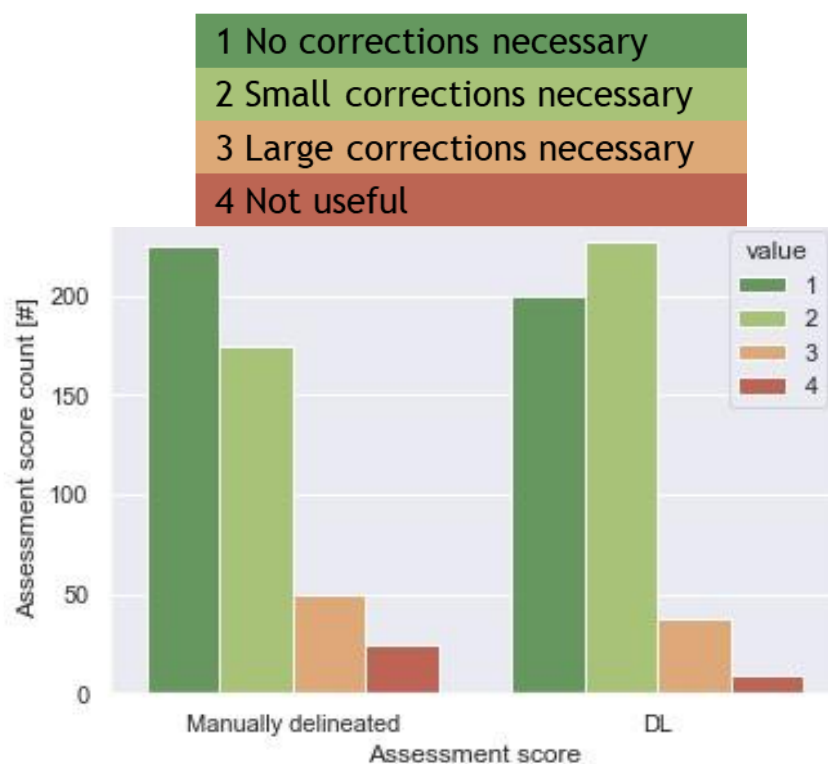
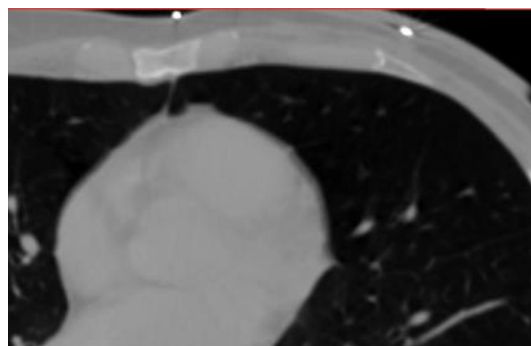
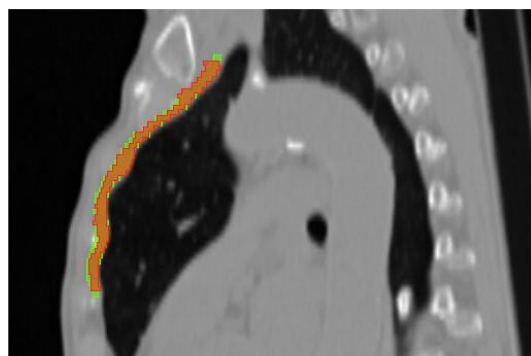
DSC(GTVt/n):
0.78/0.87



Ren et al, Acta Oncol 2021

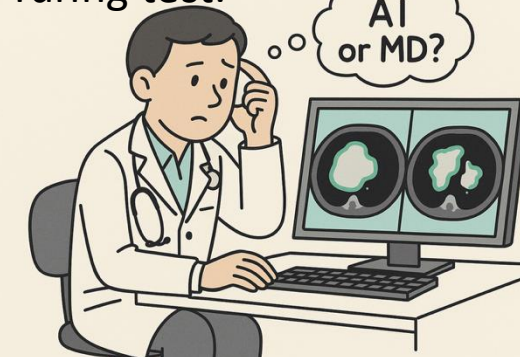
Images from Kiser et al, J Dig Imag, 2021, and generated using ChatGPT5

Qualitative metrics are employed to mimic evaluation in clinical settings

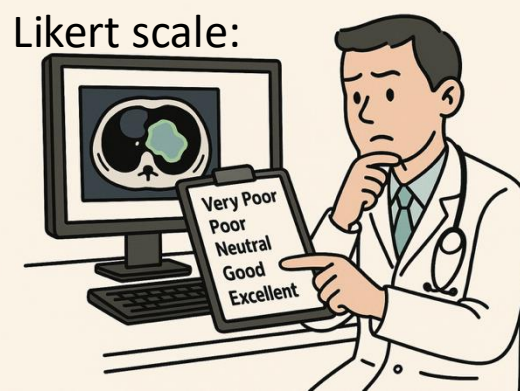


Acceptability

Turing test:

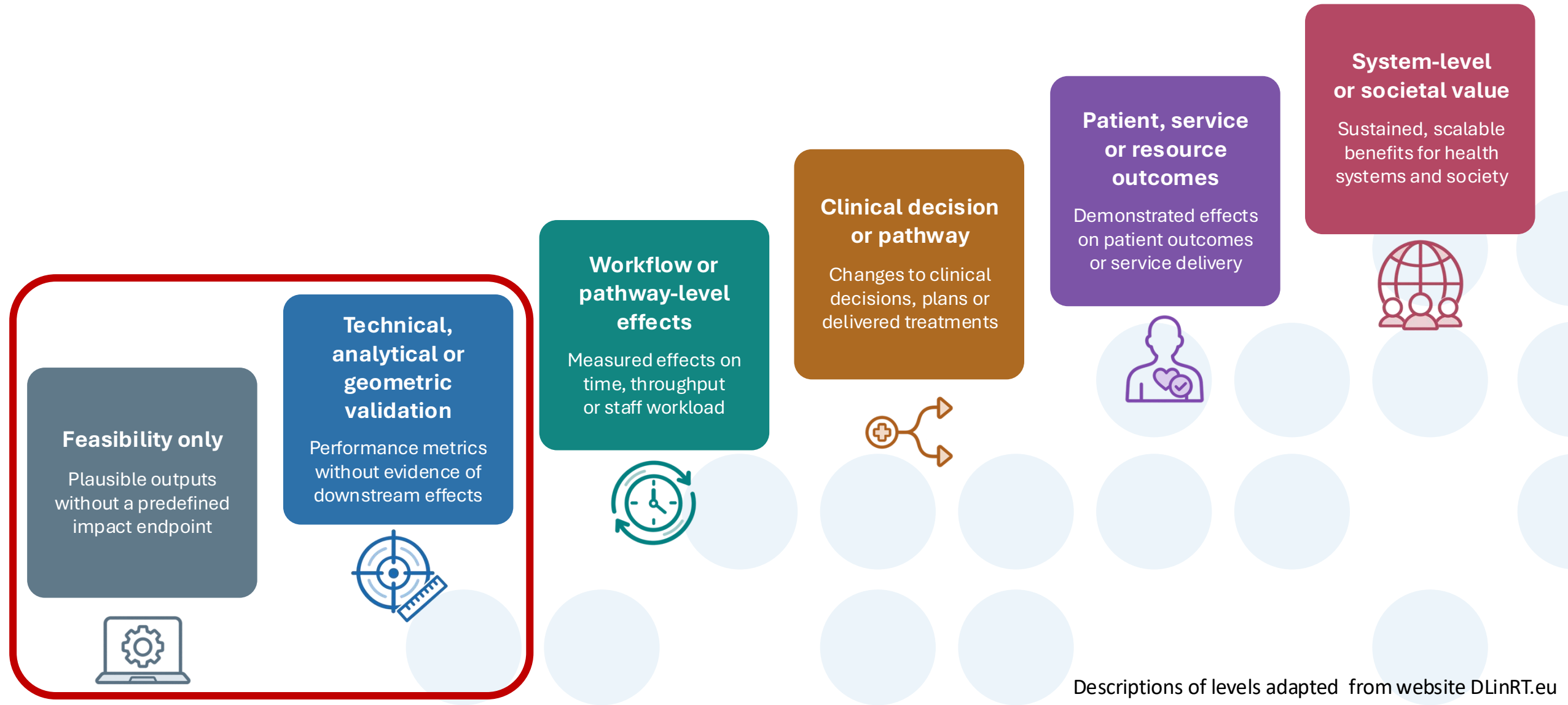


Likert scale:



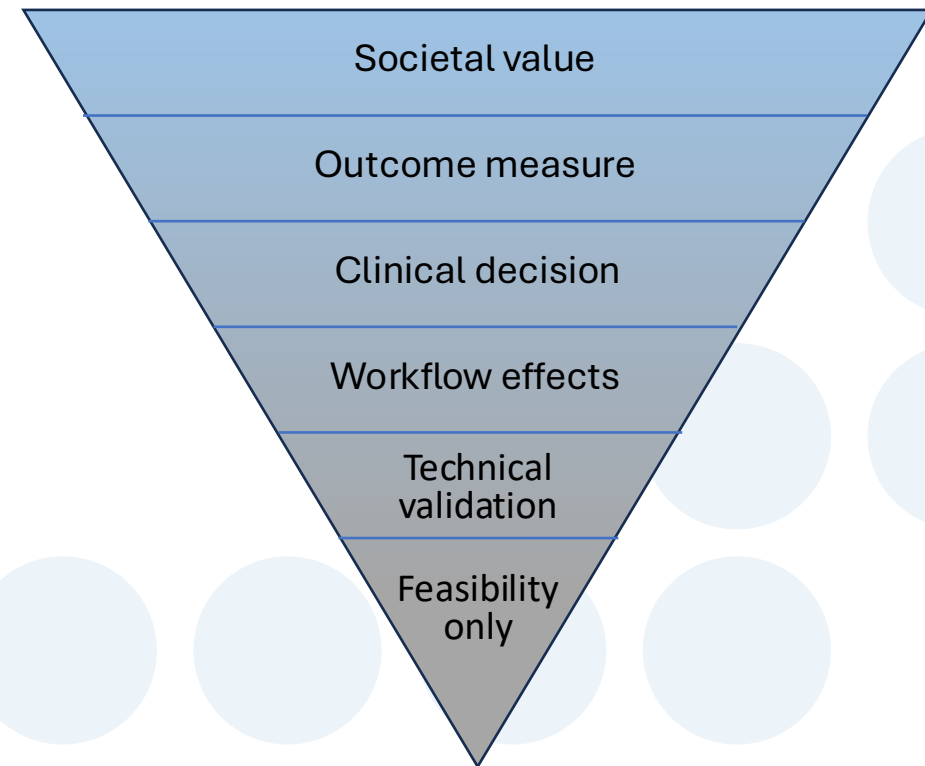
Images generated using ChatGPT5

Most AI in RT studies rank in the lowest impact categories



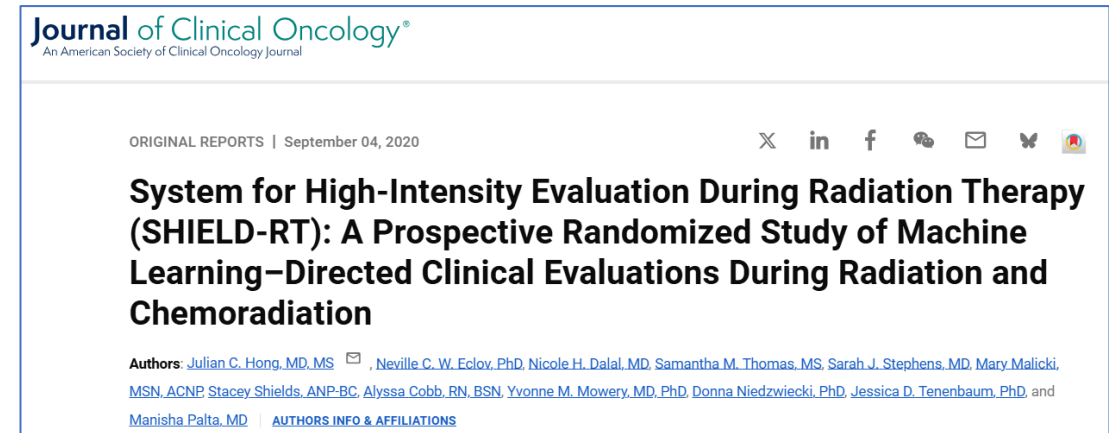
We need to build the evidence for impact – from feasibility to clinical outcome

Levels of impact:



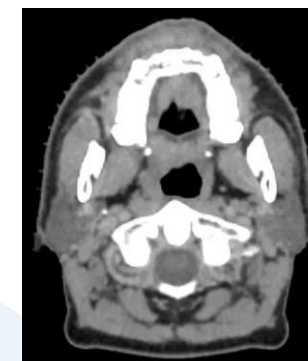
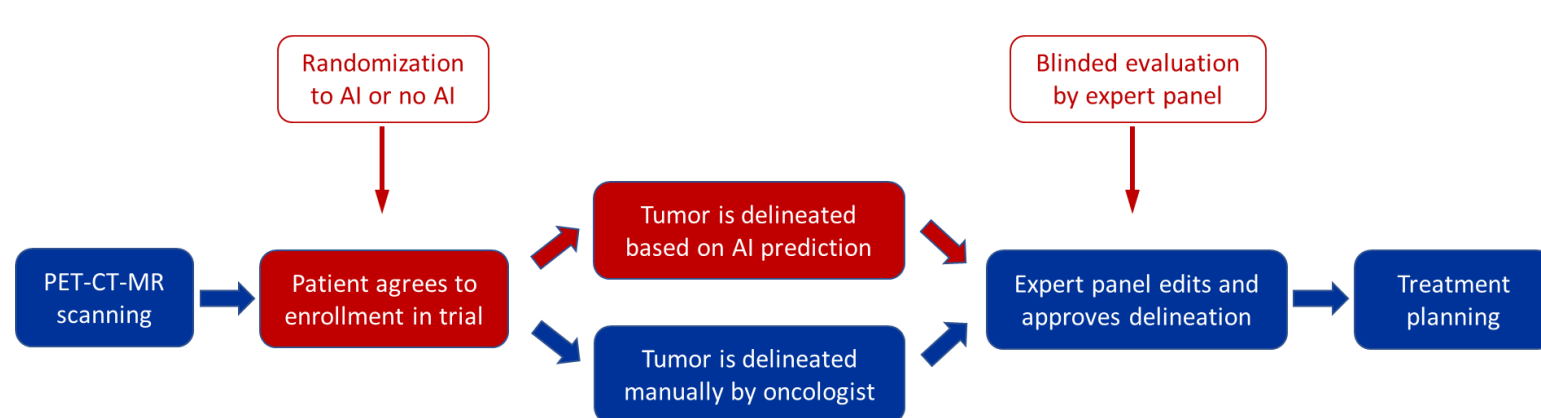
There are few high-level trials testing improved clinical outcome in RT using AI

- SHIELD-RT:
 - RCT with 963 patients
 - Machine learning accurately triaged patients directing clinical management with reduced acute care rates versus standard of care
- ASTuTE:
 - Multicentre implementation trial and prospective registry collection of real world data – goal 800 patients
 - Multimodal AI biomarkers for prognostic and predictive outputs, guiding short-term androgen deprivation therapy



A recent high-level trial tests workflow and decision impact in RT using AI

PROSA-GTV – a randomized trial of manual vs AI-assisted target delineation in head and neck cancer:

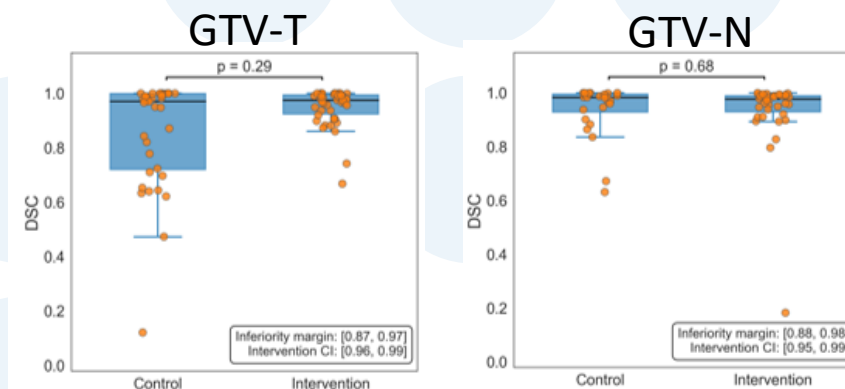


Primary endpoint:
Magnitude of changes
performed to the initial
contour at conference.

Initial
delineation

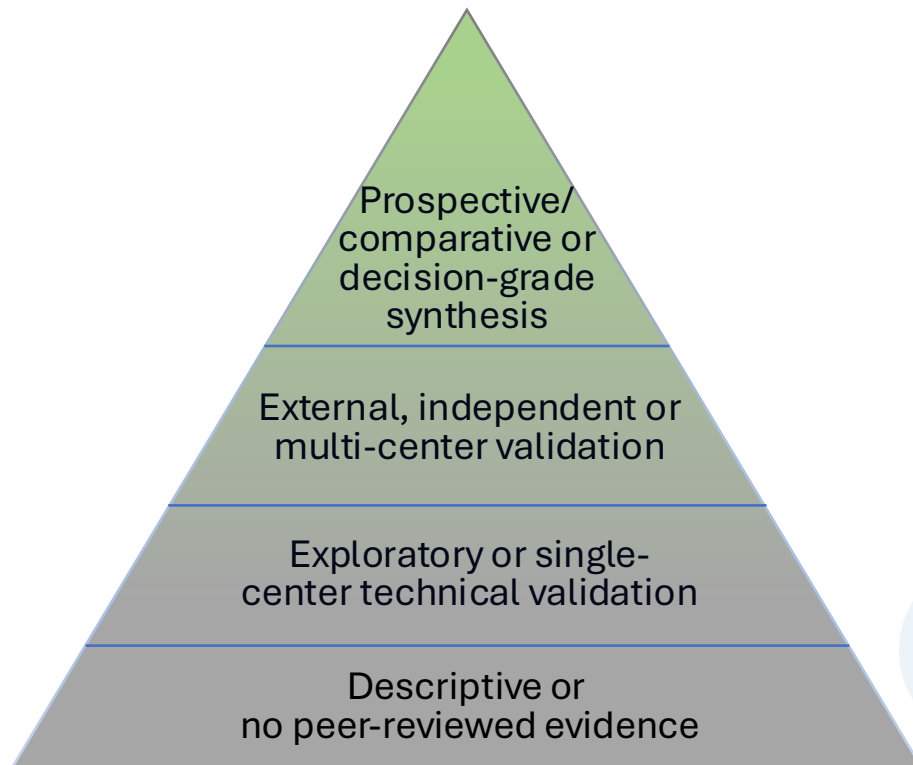


Changes at
conference

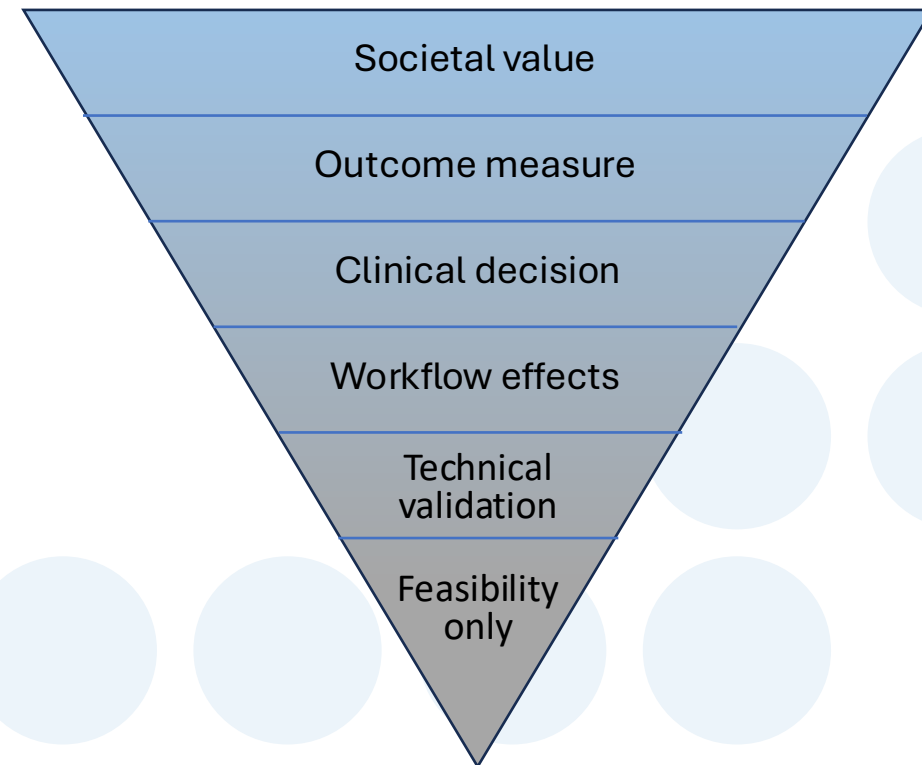


We need to build the evidence for impact – from feasibility to clinical outcome

Levels of evidence:



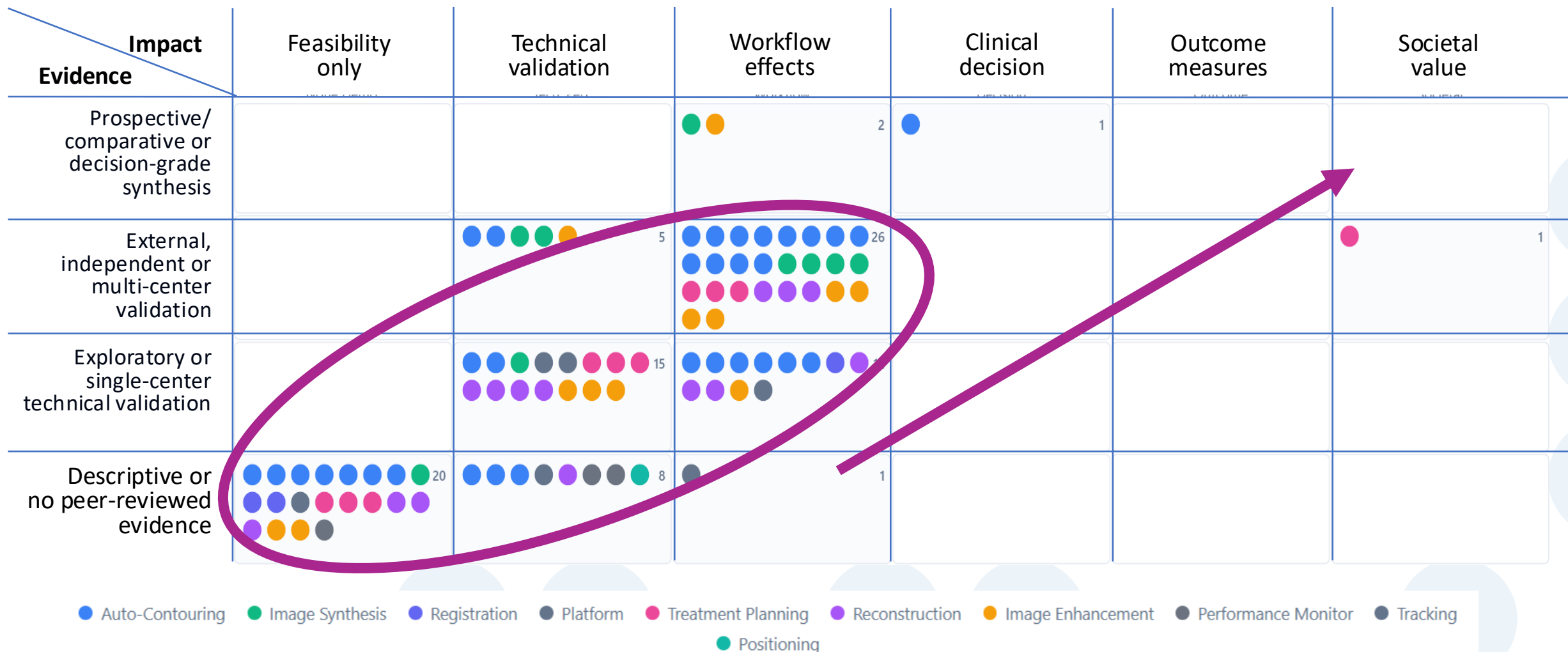
Levels of impact:



We need to build the evidence for impact – from feasibility to clinical outcome

Impact Evidence	Feasibility only	Technical validation	Workflow effects	Clinical decision	Outcome measures	Societal value
Prospective/ comparative or decision-grade synthesis						
External, independent or multi-center validation						
Exploratory or single-center technical validation						
Descriptive or no peer-reviewed evidence						

We need to build the evidence for impact – from feasibility to clinical outcome



Why are we not moving on the impact-evidence scale?

- We do not have a consensus on which levels of evidence are needed for different uses of AI
- We have not designed scalable frameworks for efficient model simulation → pilot → deployment tests
- We have a knowledge gap between disciplines hindering transferal of AI applications into meaningful clinical processes
- We direct our focus towards model development and performance, while neglecting the human factor
- Models are small and task specific – they are not scalable

Why are we not moving on the impact-evidence scale?

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Governance/policy

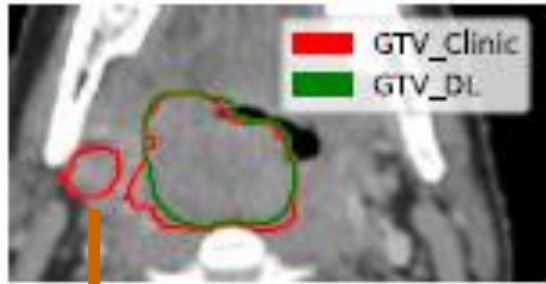
Research questions

Human perception, adoption and interaction with AI are becoming crucial factors

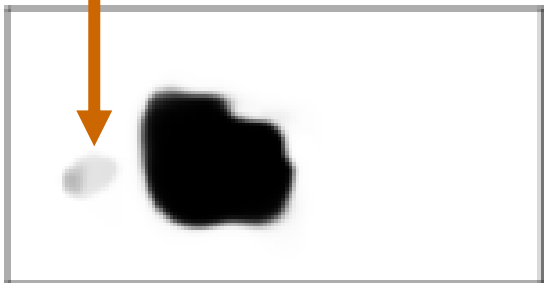
- High model performance on its own is insufficient for sustainable implementation that benefits patients and clinicians
- Human factors considerations become essential for successful clinical deployment of AI
 - Model transparency
 - Decision impact feedback
 - Interactive model prompting
 - Agentic process planning

Relevant and interesting AI research questions

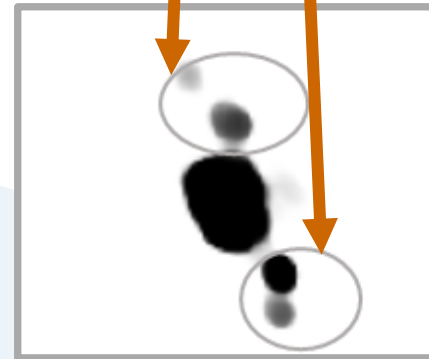
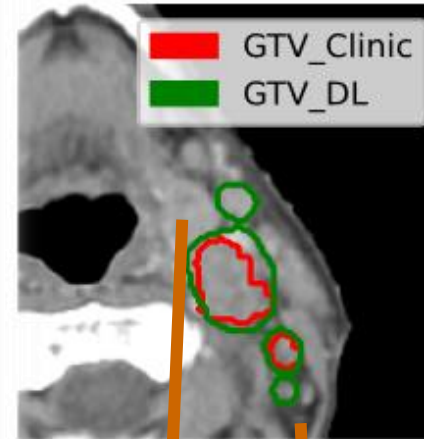
Model transparency aims to facilitate correct interpretation of model output



AI models mis-identify tumor regions due to lack of image contrast



Prediction probability map



Prediction probability map

Physics in Medicine & Biology

PAPER • OPEN ACCESS

Enhancing the reliability of deep learning-based head and neck tumour segmentation using uncertainty estimation with multi-modal images

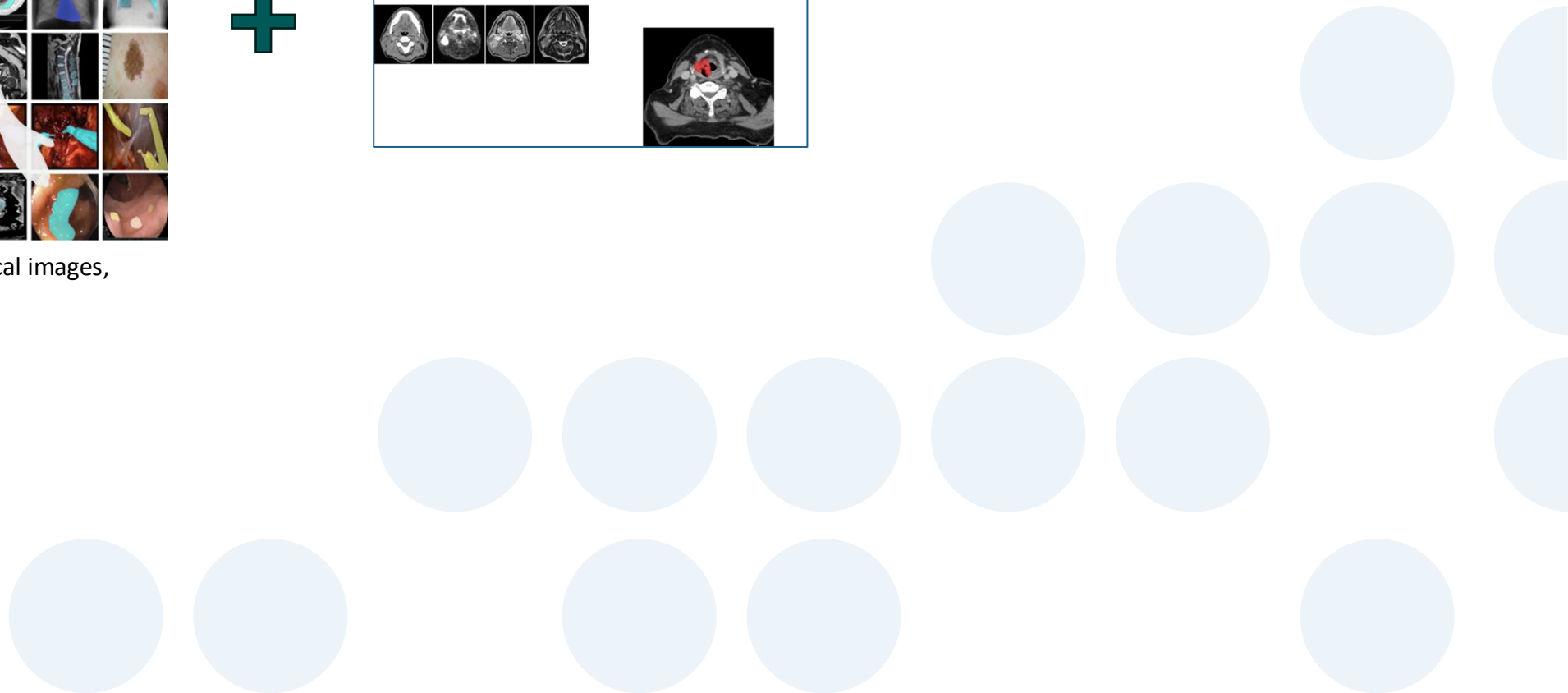
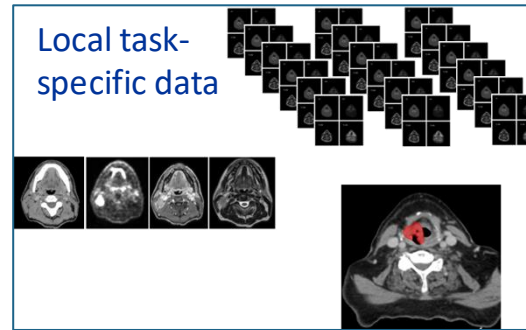
Jintao Ren*, Jonas Teuwen, Jasper Nijkamp, Mathis Rasmussen, Zeno Gouw, Jesper Grau Eriksen, Jan-Jakob Sonke and Stine Korreman

Published 5 August 2024 • © 2024 The Author(s). Published by IOP Publishing Ltd

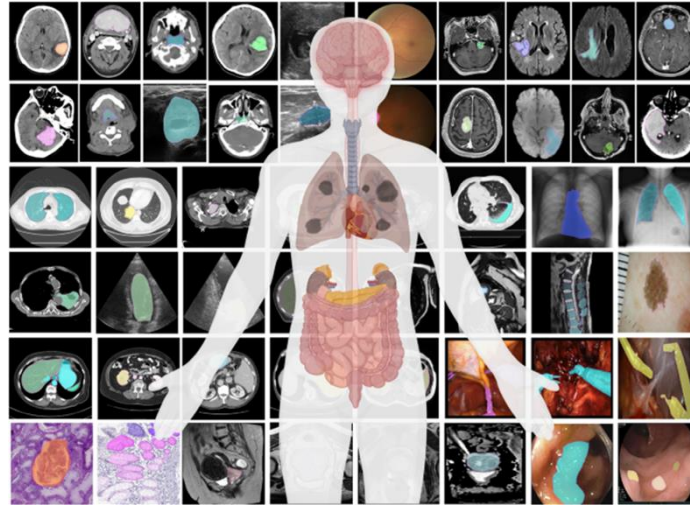
Interactive model prompting aims to include clinicians in decision making



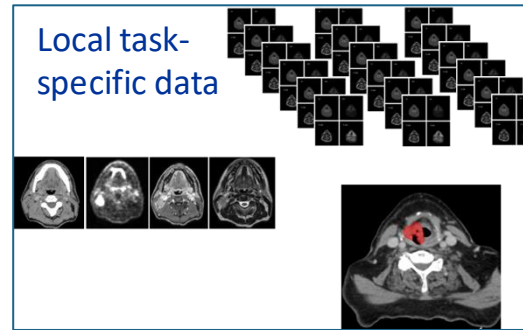
Ma et al, "Segment anything in medical images,
Nature Communications, 2024



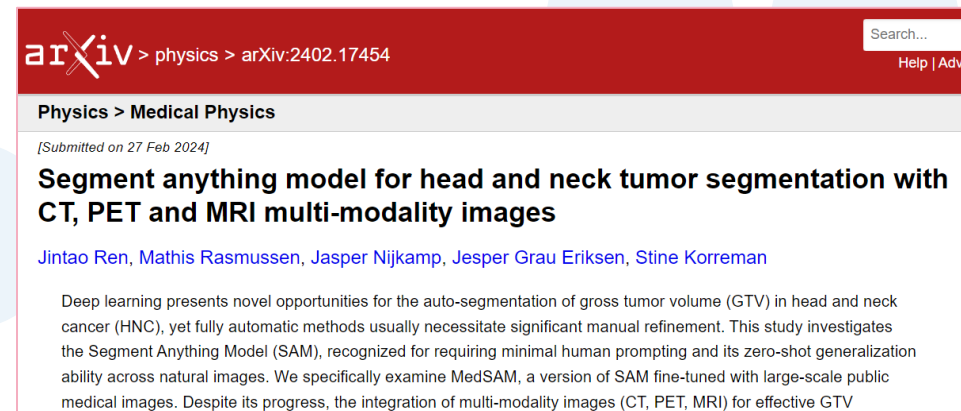
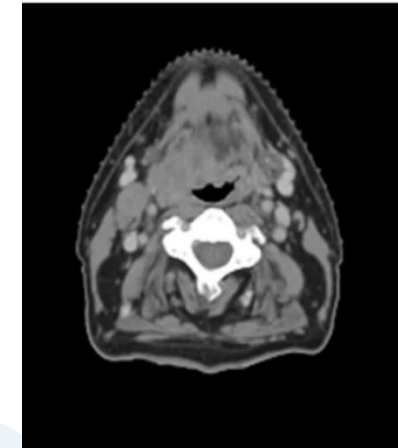
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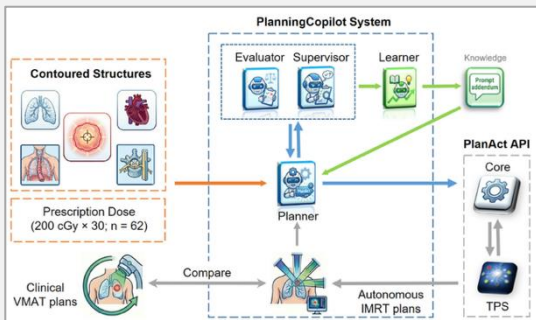


Real-time capture:



Agentic process planning aims to automate and optimize treatment decision and planning

PlanningCopilot · Mount Sinai



62 / 62

lung plans met all clinical goals; 21/21 passed measured QA

Guo H, et al. Med Phys. 2026;53:e70665

Reference-free agents · Duke

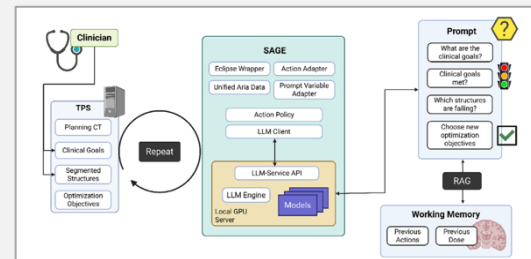


No reference plans

physician-guided, no fine-tuning; OAR sparing comparable to clinical (H&N)

Yang D, et al. Med Phys. 2026.

Local LLM · Henry Ford

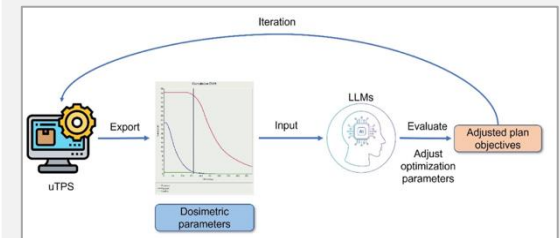


45 min

median agent compute per prostate VMAT plan; all 70B-RAG plans clinically acceptable; data stays on-site

Nusrat H, et al. Med Phys. 2026;53:e70654

Off-the-shelf LLMs · Peking Union



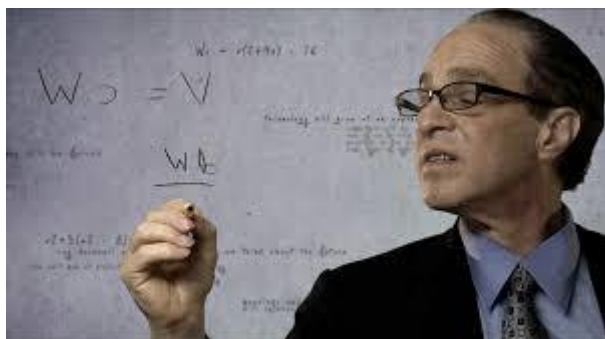
~10–16 min

per cervical VMAT plan vs ~20 min manual; OAR sparing not significantly different

Wei S, et al. Radiat Oncol. 2025;20:77

Read → Plan → Evaluate → Revise → Explain

We are closing in on "superhuman" AI ...



Ray Kurzweil
TIME Magazine 2011

1 The accelerating pace of change ...



2 ... and exponential growth in computing power ...

Computer technology, shown here climbing dramatically by powers of 10, is now progressing more each hour than it did in its entire first 90 years

COMPUTER RANKINGS

By calculations per second per \$1,000



Analytical engine
Never fully built, Charles Babbage's invention was designed to solve computational and logical problems



Colossus
The electronic computer, with 1,500 vacuum tubes, helped the British crack German codes during WW II



UNIVAC I
The first commercially marketed computer, used to tabulate the U.S. Census, occupied 943 cu. ft.

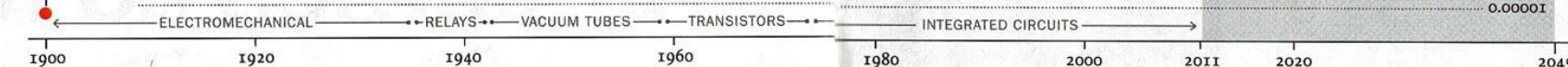


Apple II
At a price of \$1,298, the compact machine was one of the first massively popular personal computers



Power Mac G4
The first personal computer to deliver more than 1 billion floating-point operations per second

3 ... will lead to the Singularity



Deep learning

ChatGPT

2045
Surpasses brainpower equivalent to that of all human brains combined

Surpasses brainpower of human in 2023

Surpasses brainpower of mouse in 2015

Source: Time Magazine

How do we evaluate AI that may surpass human performance?

- Models already exceed human performance on narrow metrics
- Setting human performance as the 100% performance threshold will limit potential
- Autonomous, generalized AI needs new evaluation framework to promote paradigm-shifting performance

EDITORIAL



Humanity's Next Medical Exam: Preparing to Evaluate Superhuman Systems

Authors: [Jack Gallifant, M.B.B.S.](#) , and [Danielle S. Bitterman, M.D.](#)  [Author Info & Affiliations](#)

Published October 23, 2025 | NEJM AI 2025;2(11) | DOI: 10.1056/AIe2501008 | [VOL. 2 NO. 11](#)

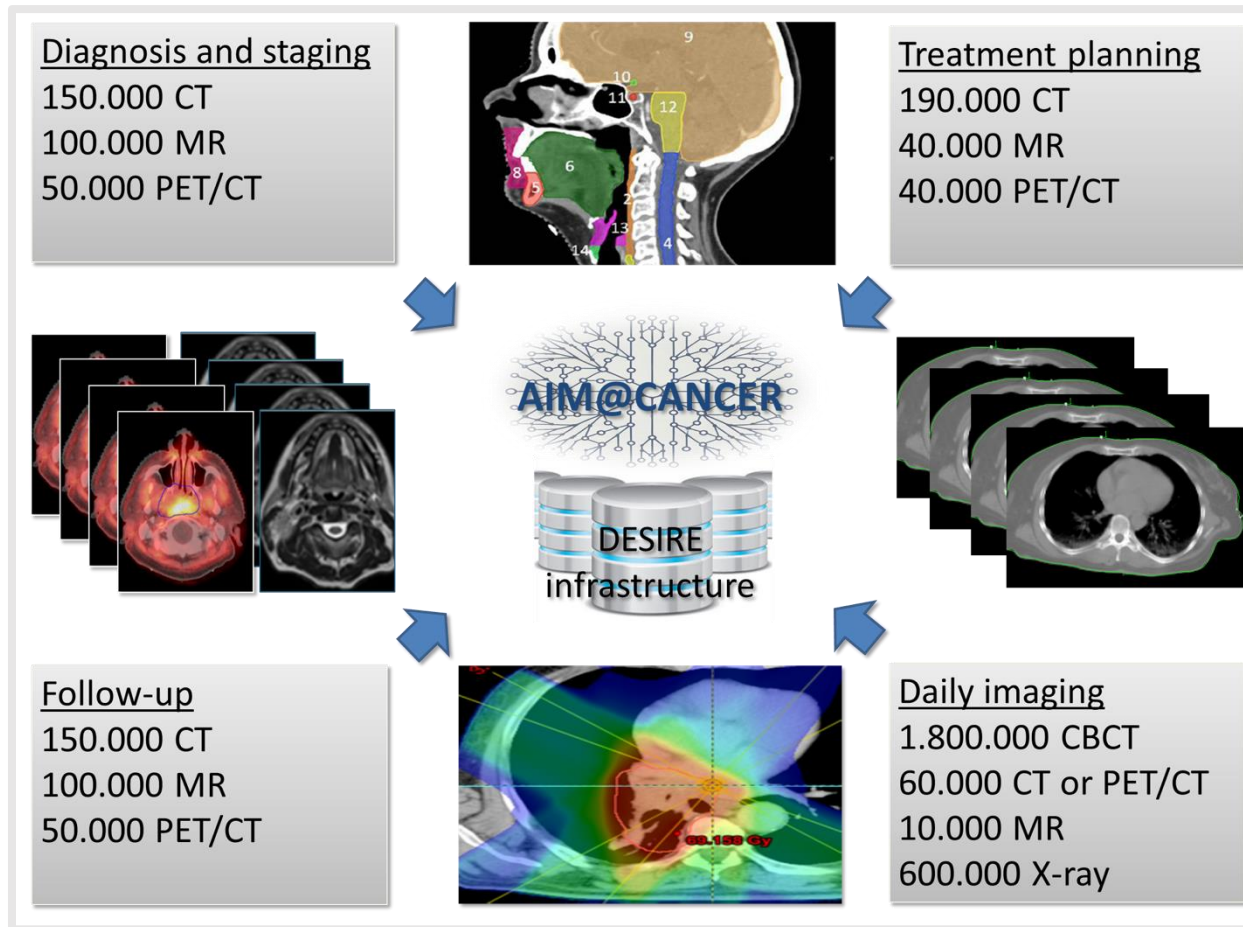
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Large models have distinct advantages but also pose challenges

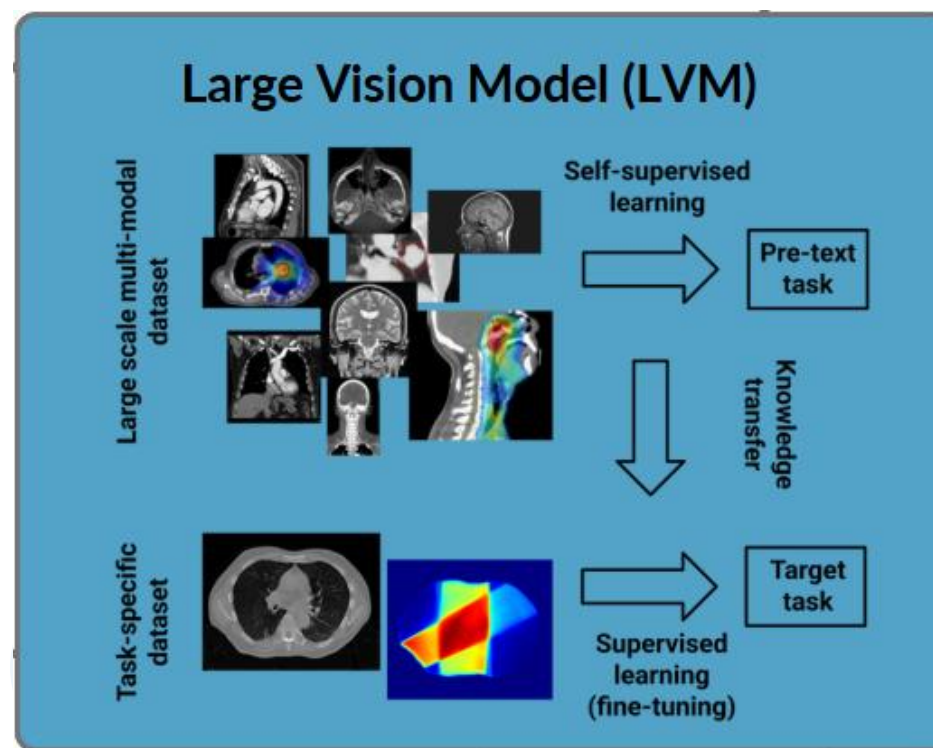
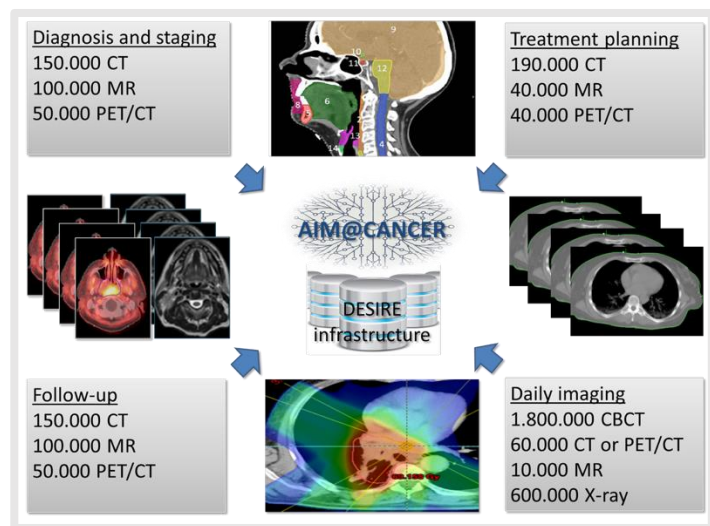
Large models (LLMs, LVMs, multimodal):

- Can exhibit “superhuman” performance
 - Can be interactively prompted
 - Are general purpose and robust
-
- Rely on large amounts of data – often “wild” data
 - Can be difficult to evaluate
 - Require large resources for training and running

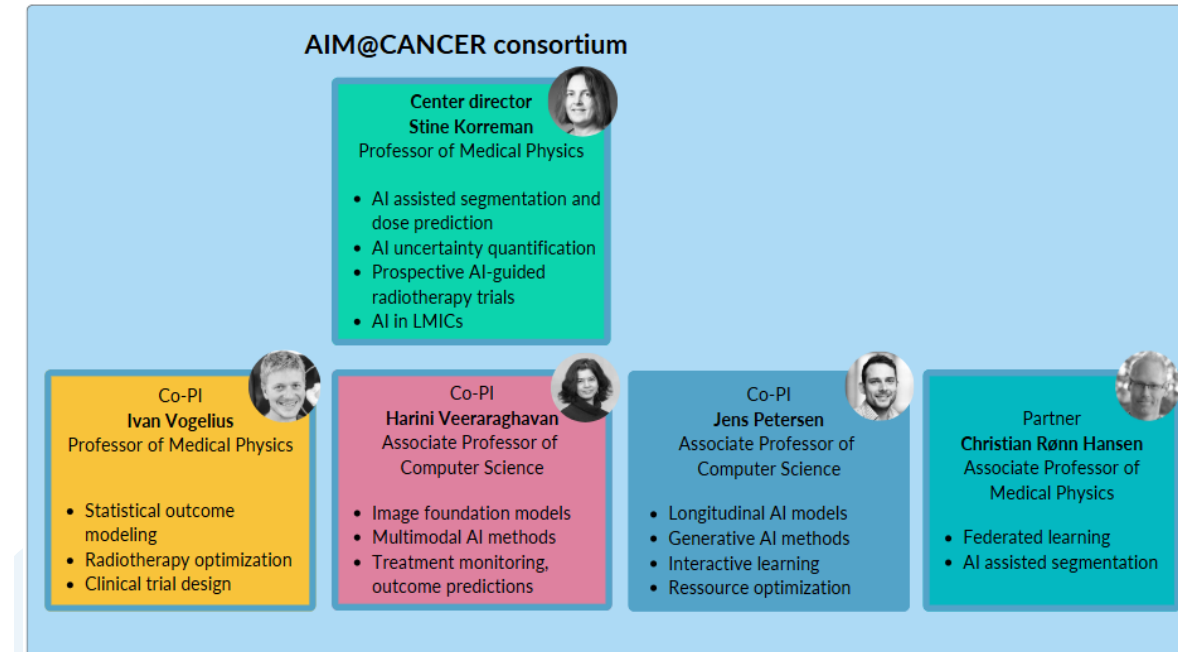
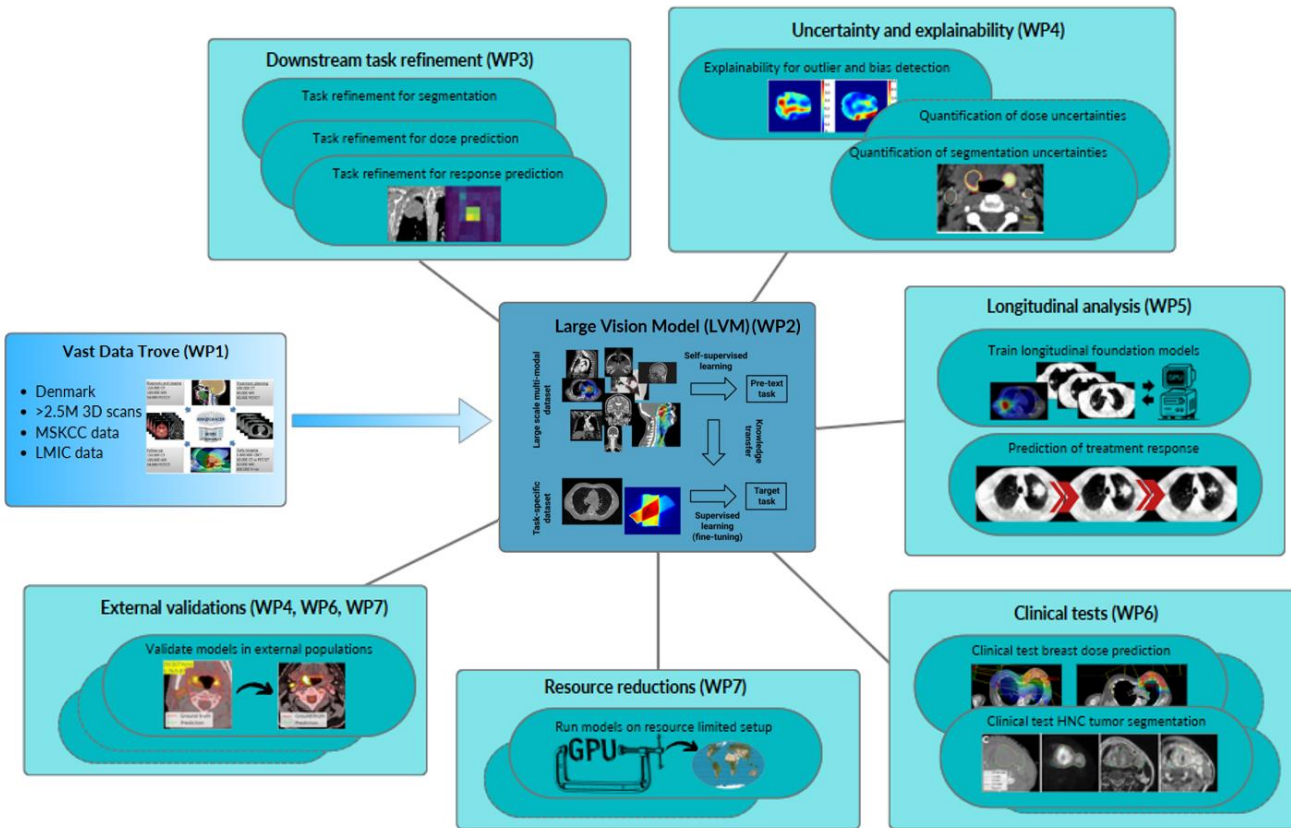
Can we make an RT domain specific large model trained on controlled data?



The AIM@CANCER research center aims to build a large population bespoke model for radiotherapy



With AIM@CANCER, we aim to generate evidence for clinical impact of large models in RT



Evidence for *AI* in radiotherapy



AI for Evidence in radiotherapy



AI-based autocontouring for reduction in inter-institutional dose variations

Physics and Imaging in Radiation Oncology 38 (2026) 100949



ELSEVIER

Contents lists available at ScienceDirect

Physics and Imaging in Radiation Oncology

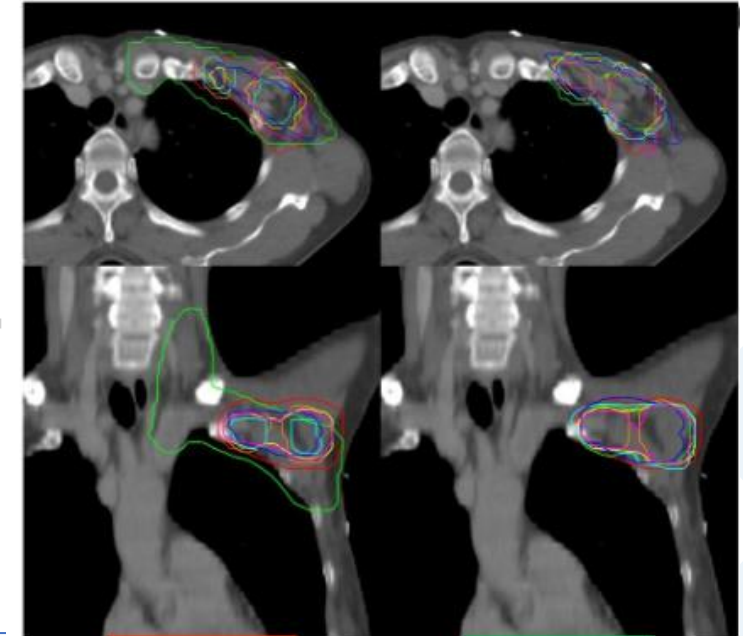
journal homepage: www.sciencedirect.com/journal/physics-and-imaging-in-radiation-oncology



Original Research Article

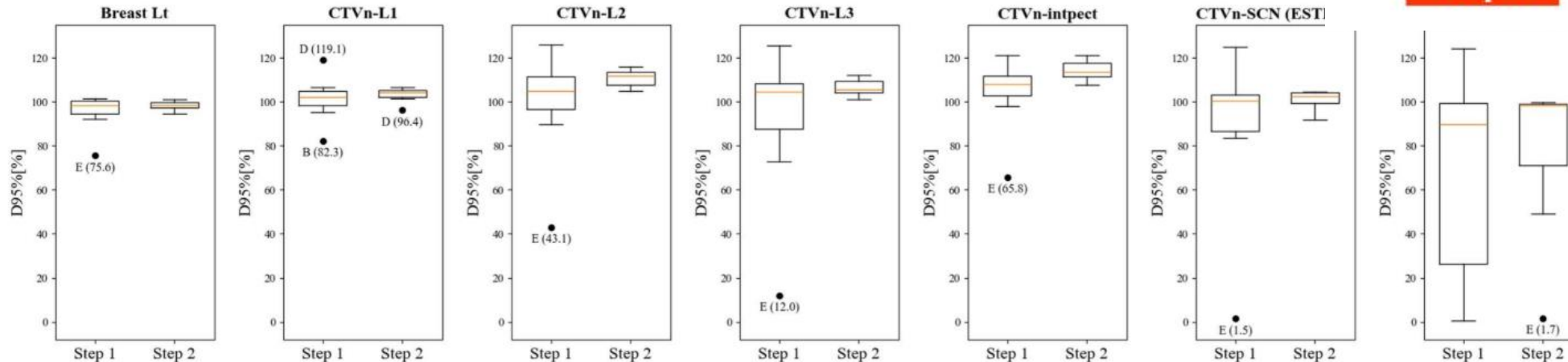
Quality assurance incorporating artificial intelligence-generated reference contours in a phase II radiotherapy trial

Wonhyeong Lee^{a,b}, Yeon-Joo Kim^{a,*}, Jin Hee Kim^c, Sung-Ja Ahn^d, Jong Hoon Lee^e,
Youngee Park^f, Jin Hwa Choi^g, Jin-ho Song^h, Yoonsun Chung^{b,*}

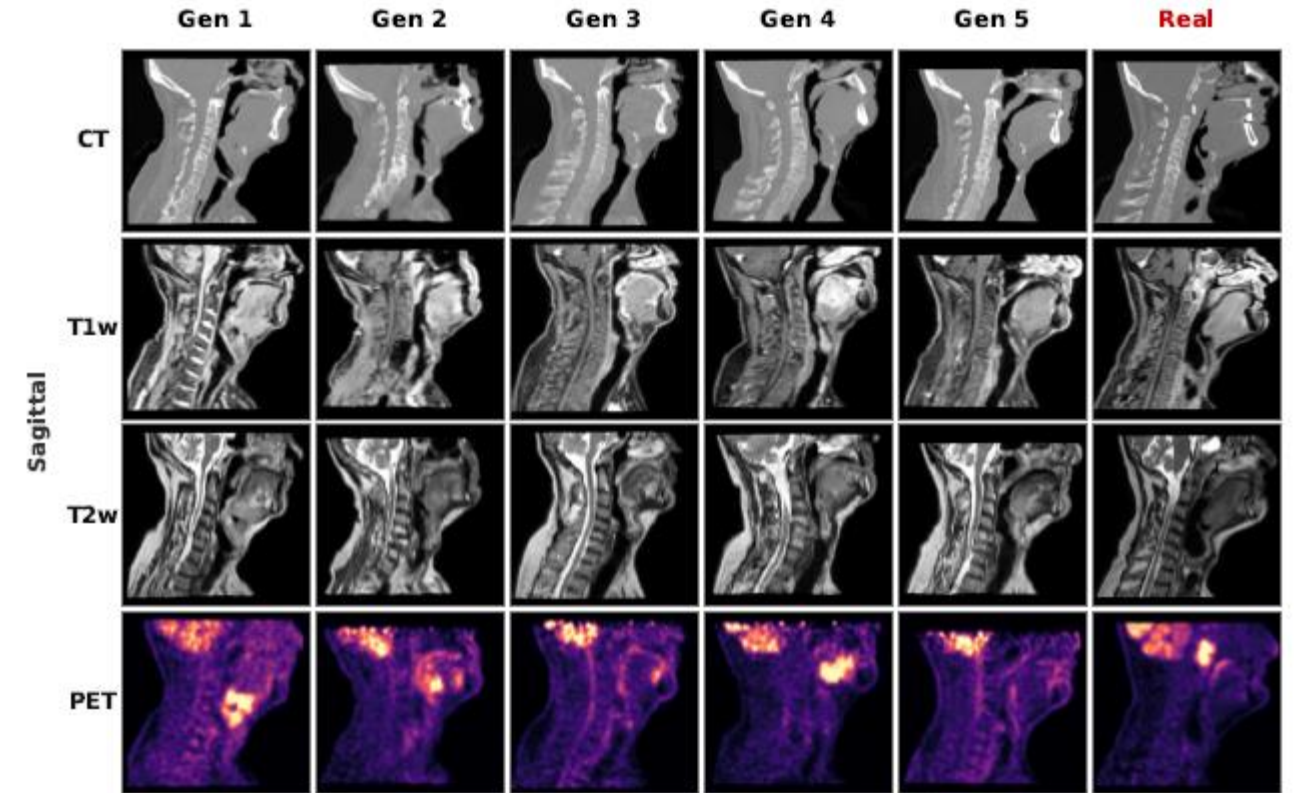
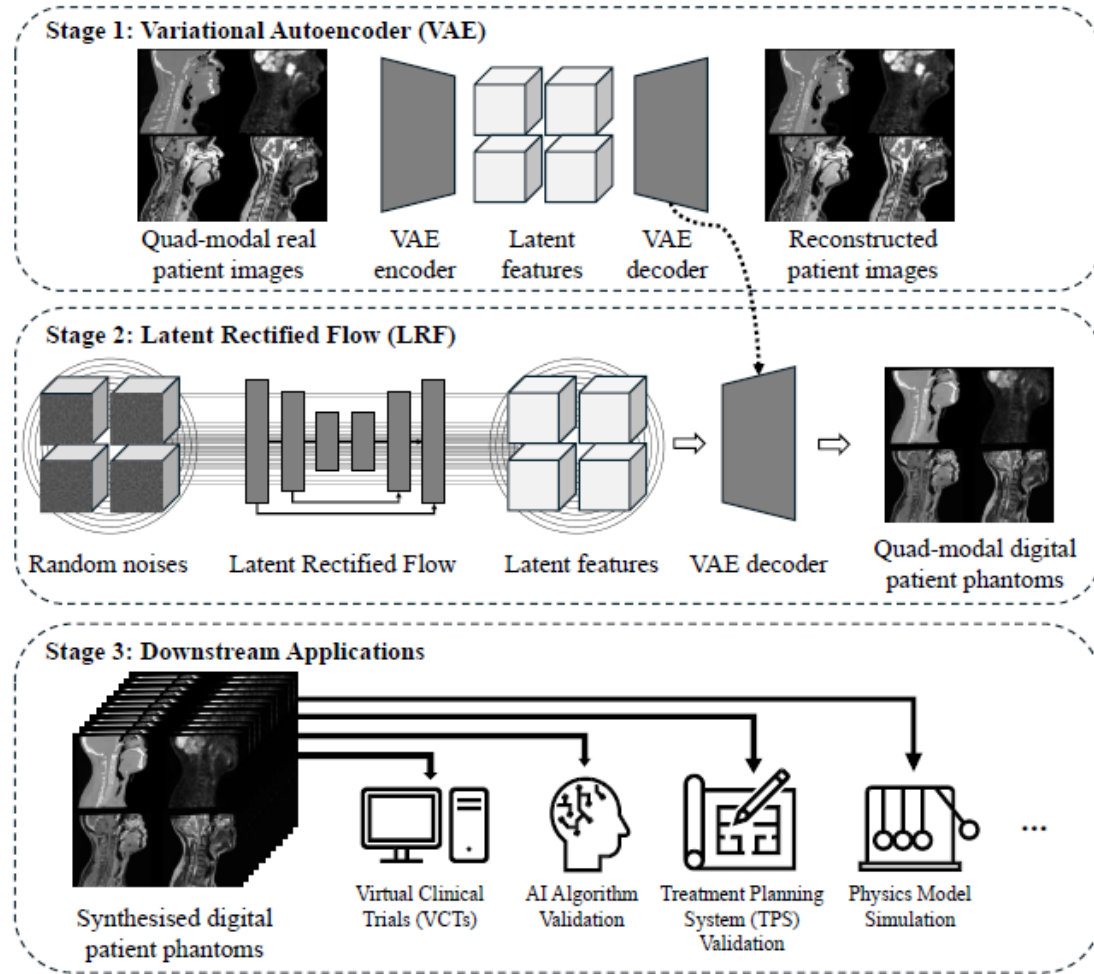


Step 1

Step 2



Virtual clinical trial concept facilitated through generation of synthetic data sets



Take-home messages

- **Clinical evidence for impact of AI in radiotherapy remains limited** – particularly for impact on patient outcomes.
- **Most current evidence concerns workflow and implementation**, while prospective clinical trials are beginning to emerge.
- **We need to focus on human factors in AI implementation** as an interdisciplinary spectrum of research.
- **Large generative AI models pose particular challenges**, that we need to address including data provenance and superhuman performance.
- **AI can enable clinical evidence generation** – through harmonization, scalable RTQA and analysis of clinical-trial data.

Thank you!



Emma Skarsø Buhl Mathis Rasmussen Jintao Ren Jesper Eriksen Jasper Nijkamp Anne IS Holm Jesper Kallehauge Birgitte Offersen Lasse Refsgaard Zixiang Wei



Kræftens Bekæmpelse



DCCC
Radiotherapy



Danish Comprehensive Cancer Center